MID-TERM EXAMINATION
(CBCS)
B. Tech CSE AI, Semester: 5th
(Oct, 2022)

Subject Code: BAI 301 **Subject: Machine Learning**
Time : 1 ½ Hours **Maximum Marks : 30**
Note:Q. 1 is compulsory. Attempt any two questions from the rest.

Q1 **(2.5×4)**

(a)Compute F1 score, TPR, TNR, FPR, and FNR from confusion matrix.

n = 165	Predicted: No	Predicted: Yes
Actual: No	50	10
Actual: Yes	5	100

(b) Create least square regression line for set of points (1,2),(-3,5).
(c) What is class conditional independence wrt probabilistic classifier?
(d) Explain the Bias-Variance tradeoff when applied to classification.

Q2 **(8,2)**

(a) W-Z are subjects, marks (0-10), A-C are grades for students P1-P6.

	W	X	Y	Z	Grade
P1	1	5	6	3	C
P2	0	9	10	5	B
P3	8	10	0	8	A
P4	9	7	9	5	A
P5	10	0	0	6	C
P6	3	7	0	9	B

Use train-test split only; apply lazy learner classifier, use 2/3rd train data.
(b)Find test accuracy of the classifier used in (a).

Q3 **(8,2)**

(a)Using the training dataset in Table, demonstrate choice of root node only using either ID3 or CART algorithm. Table 1

Day	Outlook	Temp.	Humidity	Wind	Decision
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

(b)Draw similarities and dissimilarities in the two decision trees in (a).

Q4 **(5,5)**

(a)Contrast Lasso and Ridge regression.
(b) Illustrate maximum margin hyper plane. Do applicability analysis of support vector classifier, with suitable example.